

Examiner reports

Q1.

Most candidates scored full marks in parts (a)(i) & (ii); the latter usually from the binomial formula. Answers to part (a)(iii) were less impressive. Whilst many candidates had basically the correct idea, attempts of the form $P(R \leq 15) - P(R > 5)$ or $P(R > 5) - P(R \leq 14)$ were not uncommon and a very few candidates attempted repeated use of the formula. Answers to part (b) were again disappointing. Many answers were based solely on Sly's calculated values. When the value for the binomial mean was stated, it was often as a result of intuition rather than the formula. On the somewhat rare occasions when the value for the binomial variance was found, comparisons of means and variances were done in isolation and an overall conclusion was lacking for the final mark.

Q2.

This question proved a good source of marks for many candidates with the more able often gaining at least 11 marks. Answers to part (a)(i) were often correct as most

candidates realised that $7.5 \left(\frac{15}{2} \right)$ required $P(K \leq 1)$ or $P(K < 8)$ and so scored full marks from tables. A small minority of candidates used $P(K = 7)$ or $\frac{P(K \leq 7) + P(K \leq 8)}{2}$. As in previous papers, part (a)(ii) continued to cause more difficulties almost always through use of $P(K \leq 3)$ and/or $P(K < 7)$. Knowledge of (an available source of) the relevant formulae for part (b)(i) was much improved with many fully correct answers as a result. When marks were lost, it was invariably for failing to find $\sqrt{3.6}$ for the standard deviation. Not surprisingly, only the very weakest candidates failed to score the 2 marks in part (b)(ii). In part (b)(iii), able and well-prepared candidates compared the two means (equal) and two standard deviations (different) but then often failed to commit to the overall conclusion of 'doubtful validity'. A small minority of candidates only compared $\frac{\bar{x}}{15} = \frac{6}{15} = 0.4$ with the value quoted (1 mark) or presented a qualitative argument as to the benefits of coaching (0 marks).

Q3.

This question proved a good source of marks for many candidates with the more able often gaining at least 15 marks. Answers to part (a)(i) were almost always correct and

usually by use of the formula. In part (a)(ii), most candidates realised that $7.5 \left(\frac{15}{2} \right)$ required $P(K \leq 7)$ or $P(K < 8)$ and so scored full marks from tables. A small minority of candidates used $P(K = 7)$ or $\frac{P(K \leq 7) + P(K \leq 8)}{2}$. As in previous papers, part (a)(iii) continued to cause more difficulties almost always through use of $P(K \leq 3)$ and/or $P(K < 7)$. Knowledge of (an available source of) the relevant formulae for part (b)(i) was much improved with many fully correct answers as a result. When marks were lost, it was invariably for failing to find $\sqrt{3.6}$ for the standard deviation. Not surprisingly, only the very weakest candidates failed to score the 2 marks in part (b)(ii). In part (b)(iii), able and

well-prepared candidates compared the two means (equal) and two standard deviations (different) but then often failed to commit to the overall conclusion of 'doubtful validity'. A

minority of candidates only compared $\frac{\bar{x}}{15} = \frac{6}{15} = 0.4$ with the value quoted (1 15 15 mark) or presented a qualitative argument as to the benefits of coaching (0 marks).

Q4.

Answers to part (a)(i), obtained directly from the cumulative binomial tables, were usually correct. Part (a)(ii) caused many candidates all sorts of difficulties. Whilst most candidates tried to use the aforementioned tables, many could not deal correctly with both inequalities. Thus $P(X \leq 10)$ was often used instead of $P(X \leq 9)$ and/or $P(X \leq 4)$ instead of $P(X \leq 5)$. The alternative method of using binomial formulae usually involved correct expressions but some candidates then failed to obtain a sufficiently accurate answer. Again in part (b), binomial expressions were generally correct. A small number of candidates used $n = 7$ instead of 28 or $p = 0.2$ (0.5×0.4) instead of 0.4, whilst others gave an answer to less than three significant figures. It was pleasing to see many candidates scoring the mark in part (c) for indicating that days per month vary.

Q5.

Parts (a) and (b) were considered to be routine tasks and it was disappointing to see that some candidates appeared unable to make any attempt.

Answers to part (a)(i) were usually correct, although a small minority used $p = 0.04$, rather than 0.4, in the cumulative binomial tables. Part (a)(ii), caused candidates all sorts of difficulties. Whilst most candidates tried to use the aforementioned tables, many could not deal correctly with both inequalities. Thus $P(X \leq 5)$ was often used instead of $P(X \leq 4)$ and, although many candidates found $P(X \leq 1)$ correctly, they then used this to find $P(X > 1)$ and subsequently $P(X \leq 5) - P(X > 1)$ as their final answer. The alternative method of using binomial formulae usually involved correct expressions but many candidates failed to evaluate them correctly.

Again, in part (b), binomial expressions were generally correct, although some candidates used $n = 7$, instead of 28, whilst others gave an answer to less than three significant figures.

Surprisingly, many candidates omitted part (c), this despite the specific reference to the topic in the Specification and the provision of relevant formulae in the supplied booklet. Some candidates, using the formulae, gave the value of the variance rather than the required standard deviation. Other candidates were able to intuitively state the mean correctly without explicit knowledge of its formula.

In part (d)(i), some candidates ignored the frequencies. Otherwise, most candidates used the statistical functions on their calculators with most obtaining both the mean and standard deviation correctly, although a minority reversed values and frequencies. The few candidates who had a complete set of answers to parts (c) and (d)(i) for comparison, were generally reluctant to commit to a conclusion and were often confused in their interpretations. Some candidates who failed to answer part (c) essentially made use of $\text{mean} = np$ by showing that their mean in part (d)(i) when divided by 7 gave the stated probability of 0.4.

Q6.

Answers to part (a)(i) revealed that some candidates apparently had no knowledge of the

$$\binom{17}{2}$$

binomial distribution. Of those who did, a surprising number could not evaluate correctly. However, many candidates were more successful in part (a)(ii) when they opted to use cumulative binomial tables. Attempts using the formula were less successful due to numerical inaccuracies and/or the omission of $P(X = 0)$. Part (b) generally proved beyond all but the most able candidates. The most common approach was to state $P(Y \geq 30) = 1 - P(Y \leq 29)$ and then simply switch to $p = 0.45$ in order to use Table 2.

Q7.

Answers to parts (a)(i) & (ii), using the binomial formula and cumulative tables respectively, were in the main of a good standard although in some scripts it was surprising to see poor attempts at part (i) followed by correct attempts at part (ii). Part (b) generally proved beyond all but the most able candidates. The most common approach was to state $P(Y \geq 30) = 1 - P(Y \leq 29)$ and then simply switch to $p = 0.45$ in order to use Table 2. Some better candidates who reached $P(Y \leq 20 \mid p = 0.45)$ then changed the correct answer of 0.2862 to 0.7138. Weaker candidates also often made no attempt at part (c), this despite the fact that it offered relatively easy marks. Those candidates who answered part (i) correctly, sometimes stated the variance as their answer to part (ii) or used $n = 10$. Many candidates, even with full marks for parts (i) & (ii), simply accepted the model in part (ii) because of 'two outcomes' and 'fixed trials'. Rarely did candidates comment on the discrepancy between the two values for the standard deviation.

Q8.

Most candidates answered part (a) correctly. In part (b), the considerable number of candidates who used a normal approximation did not score any marks on this part. They should be made aware of the difference between 'determine' and 'estimate'. Switching to the complementary event, in order to make use of the cumulative binomial tables, caused difficulties for many candidates. However, even with inaccurate inequalities, candidates could score three out of the four marks. Those candidates, about 40%, who calculated individual terms using the binomial formula often scored two method marks for evaluating between 6 and 8 terms but scoring the remaining two marks for a correct answer proved much more of a challenge.

Q9.

This question proved to be the best source of marks with the great majority of candidates scoring full marks. Candidates realised at last that in part (a), a description of the type of distribution (Geometric) and the value of the parameter ($p = 0.8$) were both required. Most of any errors seen in the remainder of the question were usually due to the candidates' inability to use their calculators correctly!

Q10.

Better candidates scored most, often all, of the first ten marks. Even the weakest candidates usually achieved at least three marks.

In part (a)(i) a small minority of candidates attempted $P(G = 22)$ for $B(400, 0.22)$. Most

candidates worked with $p = 0.15$ in part (a)(ii) with about the same number of candidates using tables and calculation.

A similar split was observed in answers to part (a)(iii). Here, those using tables often subtracted $P(R \leq 8)$ whereas those using the formula sometimes made numerical errors.

Answers to part (b) rarely scored both marks. Whilst many responses conveyed that P (yellow) was not constant, too many left it open as to which probability they were considering. Only the very best candidates scored the mark for ' n not fixed'. Often attempts revealed confusion between trials and beads.

Q11.

As expected, almost all candidates answered part (a) correctly. However, in part (b), far too many candidates expressed $P(5 < X < 10)$ as one of $P(X = 5, 6, 7, 8, 9)$, $P(X \leq 10) - P(X \leq 5)$, $P(X \leq 10) - P(X \leq 6)$ or $P(X \leq 9) - P(X \leq 6)$ and so scored at most one mark. Given the competent handling of the normal approximation to the binomial on recent papers, it was somewhat disappointing to see the many candidates who either ignored the continuity correction or used 149.5, rather than 150.5, in part (c). Answers to part (d) were generally poor and rarely indicated that $p = 0.15$ would be unlikely for all vehicles or that the 50 successive vehicles may not be independent. The numerous suggestions that drivers of vehicles with faulty brake lights would not stop after observing the policewoman perhaps bodes ill for future accident statistics!

Q12.

Most candidates answered part (a)(i) correctly by use of the formula for $B(20, 0.48)$. In part (a)(ii), those candidates who used tables for $B(20, 0.05)$ scored the 2 marks available, whereas those who continued with the formula approach spent unnecessary time doing so, sometimes found $P(AB = 2)$, omitted $P(AB = 0)$ or made arithmetic errors. In part (a)(iii), by far the most frequent method was to use tables for $B(20, 0.35)$. However many such candidates used

$1 - P(A \leq 8)$ rather than $1 - P(A \leq 7)$. Again those candidates continuing with the formula found themselves spending a lot of time doing so, but often with eventual success.

Many candidates used $R \sim N(48, 42.24)$ in part (b). Here frequent errors were the incorrect use or non-use of the continuity correction, the use of 42.24 rather than $\sqrt{42.24}$ for σ and giving a final answer of $1 - \Phi(z)$ rather than $\Phi(z)$.

Q13.

Answers to this type of question were much improved with many candidates scoring at least nine marks out of 12. In part (a), most candidates wrote down the correct binomial expression and then evaluated it accurately, although a small number lost the final accuracy mark through premature approximation when evaluating parts of the expression.

In part (b), most candidates identified $B(50, 0.15)$ and then used Table 1 in the formulae booklet to score full marks. The few candidates who opted for summing 11 binomial terms spent valuable time usually for no reward.

The mean and variance were usually evaluated correctly in part (c) with fewer candidates than previously equating $np(1 - p)$ to σ . Only the better candidates were able to apply

both continuity corrections correctly; all too often, the values used were 30 and 40 or 29.5 and 40.5. Nevertheless, most candidates standardised their values correctly and then used Table 3 in the formulae booklet to find a 'correct' area between $\Phi(-z_1)$ and $\Phi(+z_2)$. Answers based directly upon $B(120, 0.3)$ were awarded no marks.

Q14.

Almost without exception, candidates scored well on this question, with full marks the most common score. In part (a), candidates usually obtained 0.64499 by a correct method though, on rare occasions, some valid or invalid adjustments were made retrospectively to obtain the quoted answer. The number of completely correct answers to part (b) was

pleasantly surprising. Incorrect attempts usually involved $\frac{6}{10} \times 0.645$, $X \sim B(10, 0.6)$ or $\bar{X} \sim N\left(208.5, \frac{2.5^2}{6}\right)$.

Q15.

All candidates identified that the model $B(50, 0.2)$ was appropriate. A minority of candidates opted incorrectly for a normal approximation and scored very few marks. The vast majority opted for using tables, though a minority attempted direct calculations from the formula and, although they often scored method marks, they invariably lost accuracy marks. Some of the candidates using tables, as the question intended, made the 'classic' errors of determining $P(X \leq 10)$ in part (a) and $P(X \leq 12) - P(X \leq 8)$ in part (b).

Q16.

This was another question that was, in general, well answered with more able candidates often scoring full marks. As expected, only the weakest candidates had difficulty in answering part (a). In answering part (b), most candidates used their calculators to evaluate, or attempt to evaluate, 11 individual probabilities for $B(30, 0.75)$. Very few candidates appeared to realise that a switch to $p = 0.25$ would enable them to use cumulative binomial tables. Of those who did, about half were unsuccessful due to taking the correct answer from 1 or working with 9 or 11 rather than 10. A minority of candidates attempted a normal approximation for no reward.

Q17.

This question was answered well by most candidates, who were able to score 5 out of 6 marks. Almost all candidates identified $B(20, 0.4)$ or 0.6 , though one or two used $p = 0.5$. In part (a), candidates tended to use the formula, usually with total success. However, some candidates spoilt their attempt by quoting 0.12, rather than 0.117, whilst others either doubled or squared a correct answer. In part (b), most candidates opted for using tables and then scored full marks. A minority of candidates chose to continue using the formula, some with success, but many often lost marks for omitting $P(O = 0)$ or for numerical errors. Answers using a normal approximation were not accepted. In part (c), very few candidates realised that the salient reason was that the number of trials was not fixed. Most candidates simply referred to a large sample or non-constant proportion.

Q18.

Most candidates scored full marks in part (a) by applying the binomial formulae correctly. Few candidates wasted time and effort by attempting to answer part (b) in a similar way, but chose rather to use correctly the supplied cumulative tables.

Q19.

Nearly all candidates recognised the binomial distribution in part (a) but in part (a)(ii) many were unable to translate 20 or fewer invigorating cubes into 30 or more relaxing cubes.

There were many good answers to part (b) although some were able to list the necessary conditions for the binomial distribution but unable to apply them in the given context.

Q20.

Most candidates successfully identified $B(8, 0.3)$ but more than expected used the $p = 0.03$ column of the tables. There were a few excellent answers to parts (b) and (c) but, in the main, the explanations of why the binomial was not appropriate were rather shaky.

Q21.

Parts (a)(i), (a)(ii) and (c)(ii) were well answered. As usual, calculating the mean and standard deviation of a binomial distribution and of the observed data caused problems.

Q22.

Most were able to answer part (a) although a few incorrectly used the Poisson distribution.

The fact that unit trust managers are supposed to be better at choosing investments than a child with a pin seemed unfamiliar and so the point of part (b) was usually missed.

Q23.

This was a standard question and produced some excellent answers. Rather more candidates than expected were unable to make headway past part (a).

Q24.

Most candidates were able to identify the binomial distribution and successfully use the tables. However, only the stronger candidates were able to answer part (c) by correctly combining the probabilities calculated earlier in the question.

Q25.

Most candidates found this an easy starter, although not all were able to work out what was required in part (c). A surprising number calculated $P(26)$ in part (c) - presumably because $40 \times 0.65 = 26$.

Q26.

A surprisingly large proportion of candidates were unable to carry out the calculation in part (b). It was rare for candidates who failed at this stage to score many marks later in the question.

Q27.

Nearly all candidates recognised the binomial distribution and most were able to produce correct answers. A few attempted to calculate the probabilities. They were often successful with part (b), but this was a very time consuming method for part (a).

Q28.

Part (a) was well answered but part (b) was not. A common error was to confuse the standard deviation with the variance but many omitted this part completely. Part (c)(i) provided two easy marks for those candidates who used their calculators, but many candidates carried out lengthy and often incorrect calculations.

In part (c)(ii), many candidates ignored the instruction to relate the answer to part (c)(i) and answered in general terms instead. It was common for candidates to interpret the data in part (c) as the number of minutes Siballi had to wait for a bus rather than the number of times out of ten Siballi had to wait more than 6 minutes.

Q29.

Almost all candidates recognised the binomial distribution, although in part (b) a few were unable to translate 40 purchases completed into zero purchases not completed.



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